

- Continue from last lecture discussion of Q and S.
- Section 2.4 & 2.5 of Ohring's textbook
 - pumps
 - pumping systems
- Physical vapor deposition (Chap. 3 - 5, Ohring's textbook)

We will concentrate on the following processes:

i) Evaporation: atoms/molecules are removed from the source by thermal means.

MBE
Thermal
E-beam



ii) Sputtering: atoms/molecules are removed from source through the impact from gaseous ions.

iii) Pulsed-laser deposition/ablation: atoms/molecules are vaporized from source through light energy absorption from laser

• Evaporation → deposition

- First systematic investigation of evaporation took place over 140 years ago (1882, Hertz) with the investigation of Hg distillation and evaporation.

- In 1915 investigations by Knudsen and Langmuir resulted in a familiar expression for solid evaporation

$$\bar{\Phi}_e = \alpha_e^* (P_e - P_h) \frac{N_A}{\sqrt{2\pi MRT}}$$

α_e^* $\equiv$ a coefficient that accounts for factors such as surface cleanliness and reactivity

$P_h \equiv$ hydrostatic pressure acting on evaporant.

$P_e \equiv$ equilibrium evaporant pressure

$$\therefore \bar{\Phi}_{e_{\max}} = P_e \left(\frac{N_A}{\sqrt{2\pi MRT}} \right) \quad \text{when } P_h = 0$$

$\alpha_e = 1$ (e.g., Au)

- Evaporation sources

+ Knudsen cells^(1,2) - molecular beam epitaxy (MBE)

+ Open boats or wire baskets - less controlled evaporation

$$(1) \bar{\Phi}_{KC} = 3.51 \times 10^{22} \frac{P_e}{\sqrt{MT}} \left[\frac{m_c}{\text{cm}^2 \cdot \text{s}} \right] \quad \left(\text{cgs, torr units and KC. opening in cm}^2 \right)$$

or

$$(2) \bar{\Phi}_{KC} = 5.834 \times 10^{-2} \sqrt{\frac{M}{T}} P_e \left[\frac{\text{g}}{\text{cm}^2 \cdot \text{s}} \right] \quad (M \text{ in g/mol})$$

$$(1) \bar{\Phi}_{KC} \times \frac{M}{N_A} = (2) \bar{\Phi}_{KC}$$

Evaporation $\rightarrow$ deposition (continued)

- Knudsen approach usually involves heating by thermal means
- Electron beam evaporation (3.4.2, Ohring) involves heating with an electron beam. E-beam deposits energy into the solid source to locally heat it.
 - Can lead to high-purity films
 - can be modeled as a point source

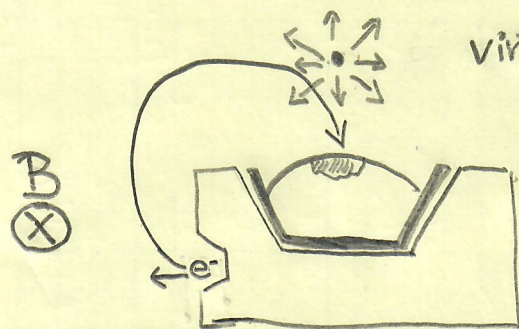


Fig. 3-14

- Can evaporate high melting-temperature materials

Evaporation $\rightarrow$ deposition (continued)

- Depending on the hardware and configuration sources can usually be thought of as point or surface sources.
- These sources are of the "line-of-sight" type.
- Deposition is usually performed under a level of vacuum.
- Atoms and molecules that arrive at the substrate are thermalize and, in the absence of substrate heating, have very little mobility
- The evaporation pressure, P_e , can be determined from the Clausius - Clapyeron relationship

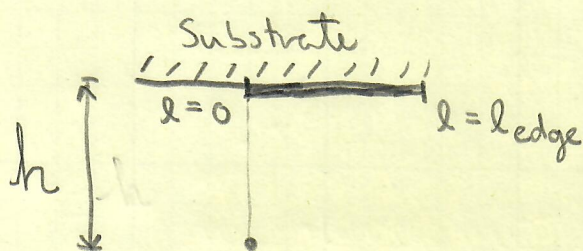
$$P_{\text{vap}} = P_0 \exp\left(-\frac{\Delta H_v}{RT}\right); P_{\text{vap}} \equiv \text{vapor pressure}$$
 where $\Delta H_v \equiv$ enthalpy of vaporization

Fig. 3-1 and Fig 3-2 present vapor pressures for elements.

$$P_e \approx P_{\text{vap}}$$

Evaporation $\rightarrow$ deposition (cont'd)

Film-thickness uniformity (3.3)

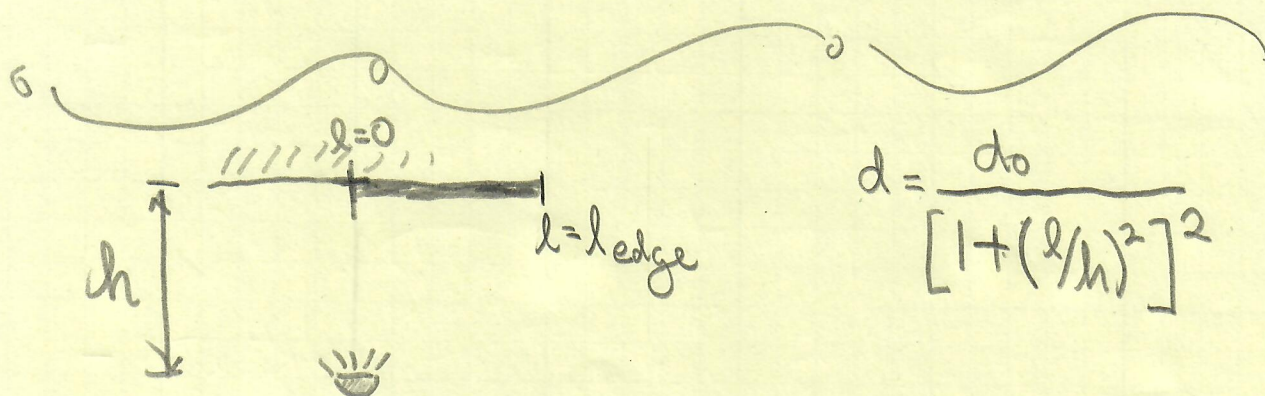


d_0 = thickness at $l=0$

d = thickness along substrate

$$d = \frac{d_0}{[1 + (l/h)^2]^{3/2}}$$

point source
(e.g., e-beam deposition)



$$d = \frac{d_0}{[1 + (l/h)^2]^2}$$

surface source

Conformality (Fig. 3-10)

- Due to the line-of-sight nature of physical vapor deposition and (relatively) low mobility of deposited atoms/molecules, shadowing effects can lead to void formation and "blind spots" as depicted in Fig 3-10.

